

KANNAN S

Senior Frontend Engineer | React.js | Next.js | TypeScript | System Design | Team Leadership

Bangalore, Karnataka, India | +91 9916674585 | kannan.skannan111@gmail.com | linkedin.com/in/kannan-s-3567471a3

PROFESSIONAL SUMMARY

Senior Frontend Engineer with 5+ years of experience architecting scalable, high-performance web applications using **React.js, Next.js, and TypeScript** across the iGaming, AI, and AR domains. Currently leading a **7-member development team** as Assistant Project Lead (APL), driving technical architecture, code quality, and engineering standards adopted organization-wide. Delivered measurable business impact including a **40% reduction in page load time**, a **35%+ improvement in Lighthouse scores**, a **60%+ reduction in unnecessary re-renders**, and a **30% decrease in production runtime errors**. Specialized in real-time systems (Socket.IO, WebSockets), Core Web Vitals optimization, micro-frontend architecture, and cross-functional engineering leadership within Agile/Scrum environments.

TECHNICAL SKILLS

Languages & Frameworks: React.js, Next.js, JavaScript (ES6+), TypeScript, HTML5, CSS3, SCSS, Three.js, React Flow

UI & Styling: Material UI, Tailwind CSS, Styled Components, Responsive Design, Design Systems

State Management & APIs: Redux Toolkit, React Query, Context API, GraphQL, REST APIs, Axios, WebSockets

Real-Time & Architecture: Socket.IO, Event-Driven Architecture, Micro-Frontends, Module Federation, Scalable System Design

Performance Engineering: Core Web Vitals (LCP, CLS, FID), Lighthouse, Code Splitting, Lazy Loading, Tree Shaking, Memoization, Bundle Analysis, Webpack, Vite, Turbopack

Leadership & Process: Team Leadership, Code Review, Technical Mentoring, Fresher Onboarding, Technical Interviews, Agile/Scrum, Coding Standards, Stakeholder Communication

DevOps & Infrastructure: Firebase, CI/CD, GitHub Actions, Deployment Pipelines, OAuth 2.0, JWT, Google Search Console

Testing & Quality: Jest, React Testing Library, ESLint, Prettier, Accessibility (WCAG/a11y), Git, Version Control

WORK EXPERIENCE

Frontend Engineer | Assistant Project Lead (APL)

Nov 2024 – Present

GammaStack — Real-Time Multiplayer Gaming Platform | Bangalore, India

- Team Leadership:** Lead a 7-member frontend engineering team, conducting code reviews, mentoring junior developers, establishing coding standards, and driving technical architecture decisions adopted platform-wide.
- Architected and shipped **50+ production-grade Next.js components** for a high-traffic real-time multiplayer gaming platform serving thousands of concurrent users, including game lobbies, matchmaking UI, live leaderboards, in-game notifications, and player progression systems.
- Performance Engineering:** Improved Lighthouse scores by **35%+** and reduced average page load time by **40%** via advanced code splitting, tree shaking, image optimization, and strategic caching, directly improving Core Web Vitals (LCP, CLS, FID) and user retention during peak traffic.
- Real-Time Architecture:** Built bidirectional real-time features using Socket.IO for live game-state synchronization, score broadcasting, and event streaming, achieving **sub-100ms UI updates** at peak concurrency with optimized message handling.
- Render Optimization:** Eliminated **60%+ of unnecessary re-renders** using advanced React.memo patterns, useMemo, and useCallback strategies, measurably improving user retention during high-traffic gaming sessions.
- SEO & Organic Growth:** Resolved critical Google Search Console indexing and crawl errors, improving technical SEO health and driving a measurable increase in organic traffic.
- Engineering Excellence:** Designed scalable frontend architecture patterns and coding standards; conducted **25+ technical interviews**; delivered a structured 4-week onboarding program for 10 freshers. Awarded 'Valuable Gem Award 2025' for platform contributions and team leadership.

Senior Executive Software Engineer (Frontend)

Sep 2021 – Oct 2024

SoftSuave Technologies — Product & Client Engineering | Bangalore, India

- Delivered **3 full-scale products** across AI, AR, and productivity domains using React.js and Next.js, owning the full lifecycle from requirement analysis and architecture through production deployment.

- **TypeScript Adoption & Code Quality:** Enforced TypeScript across all codebases, reducing production runtime errors by ~30% via strict type checking and significantly reducing new-developer onboarding time.
- **AR & 3D Technology:** Built a no-code AR instructional guide builder using Three.js and React Flow, enabling non-technical users to create 3D augmented-reality guides and reducing on-site service calls.
- **Authentication & Security:** Implemented OAuth 2.0 (Google Sign-In, Discord OAuth) with JWT-based session management across multiple product lines, improving security posture and user onboarding conversion.
- **Global Scale & Optimization:** Implemented PWA capabilities and i18n (internationalization) support, expanding product reach to **5+ geographies**. Optimized the image compression pipeline, reducing bandwidth usage by **45%**.

KEY PROJECTS & TECHNICAL ACHIEVEMENTS

Real-Time Multiplayer Gaming Platform

Tech Stack: Next.js, React.js, Socket.IO, Redux Toolkit, Tailwind CSS, REST APIs, Firebase

- Architected scalable lobby, matchmaking, leaderboard, and notification UI components serving thousands of concurrent players with sub-100ms real-time updates via WebSocket architecture.
- Achieved **40% load-time reduction** through advanced code-splitting, lazy loading, image optimization, and Core Web Vitals tuning; improved LCP by **45%**, CLS by **60%**, and FID by **50%**.

GIDR — Conversational AI Platform

Tech Stack: Next.js, TypeScript, Material UI, REST APIs, OAuth 2.0, Streaming APIs

- Led frontend delivery of an AI platform enabling users to query PDFs, videos, and call recordings; architected streaming response rendering and real-time sentiment-analysis UI.
- Integrated Google OAuth 2.0 and Discord OAuth for seamless onboarding; owned architecture decisions and API integrations across a 3-person frontend team.

CareAR Experience Builder — No-Code AR Guide Platform

Tech Stack: React.js, Three.js, React Flow, Material UI, Firebase

- Built a drag-and-drop AR guide builder with interactive Three.js 3D model rendering; delivered **200+ reusable UI components**, reducing cross-team development time by **30%**.
- Delivered enterprise-grade features used by clients to reduce on-site technician visits and improve service delivery efficiency.

AWARDS & RECOGNITION

- **Valuable Gem Award 2025** — GammaStack: Recognized for outstanding platform contributions, 35%+ performance improvements, engineering team leadership, and building the fresher onboarding program.
- **Responsible Worker Q2 2024** — SoftSuave Technologies: Awarded for exceptional ownership, zero-defect delivery, and cross-functional collaboration across 3 concurrent product teams.

EDUCATION

Bachelor of Technology — Computer Science Engineering

R.R Institute of Technology, Bangalore | Visvesvaraya Technological University (VTU) | 2016 – 2020